MATH 152 **QUIZ11** NAME _____

Mrs. Bonny Tighe 25 points

10.1,10.2,10.3 TA_____ Wed 12/6/2017

1. Sketch the plane curve defined by the given parametric equations. Indicate with an arrow the direction in which the curve is traced as the parameter increases. Eliminate the parameter to find a Cartesian equation of the curve.

a) $x = 5\sin\theta, \quad y = 4\cos\theta$

b) $x = 2 + e^{t}, \quad y = e^{2t}$

2. Find the equation of the tangent line to the curve $x = t^2, \quad y = t^3$ at (1, - 1)

3. Find the area enclosed by the x- axis and the curve $x = t^3 + 1, \quad y = 2t - t^2$

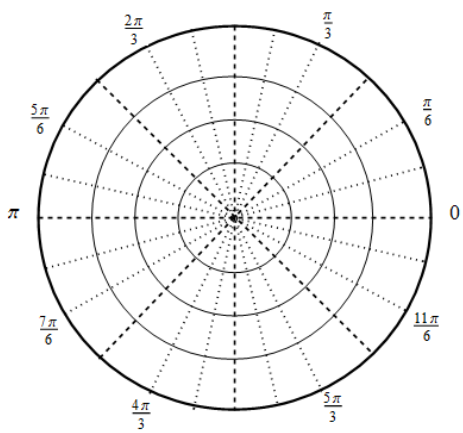
4. Compute the surface area obtained by rotating the curve

$x = e^{t} + e^{-t}, \quad y = 5 - 2t, \quad 0 \le t \le 3$ about the y-axis. $S = 2\pi \int_{\alpha}^{\beta} radius(arc\ length)$

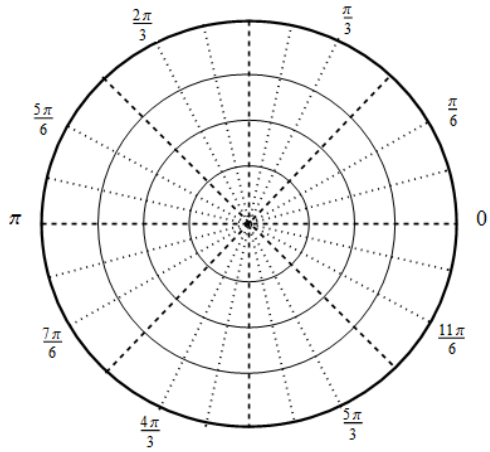
5. a) Find the Cartesian coordinates for the polar coordinate $\left(-2, 7\pi/4\right)$

b) Find polar coordinates for the Cartesian coordinate (- 4, 4) where $r < 0, and\ \theta < 0$

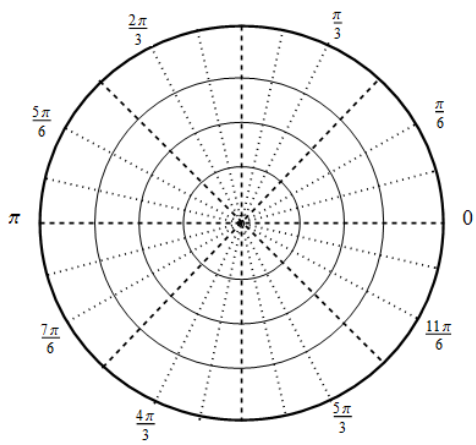
6. Sketch the graph. $r = 2 + \cos 2\theta$.



7. Sketch the graph. $r = 3 + 2\sin\theta$



8. Sketch the graph. $r = 2 + 3\sin\theta$



MATH 152 **QUIZ 11 A** NAME _____

Mrs. Bonny Tighe 25 points

10.1,10.2,10.3 TA_____ Wed 12/6/2017

1. Sketch the plane curve defined by the given parametric equations. Indicate with an arrow the direction in which the curve is traced as the parameter increases. Eliminate the parameter to find a Cartesian equation of the curve.

a) $x = 3\cos\theta,\ \ y = 4\sin\theta$

b) $x = e^{t} - 3,\ \ y = e^{2t}$

2. Find the equation of the tangent line to the curve $x = t^{2} + 1,\ \ y = t^{3}$ at (2, - 1)

3. Find the area enclosed by the x- axis and the curve $x = t^{3} + 1,\ \ y = 3t - t^{2}$

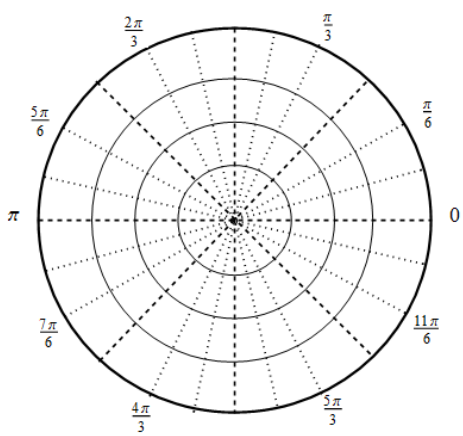
4. Compute the surface area obtained by rotating the curve

$x = e^{t} + e^{-t}, \quad y = 3 + 2t \ , \quad 0 \le t \le 2$ about the y-axis. $S = 2\pi \int_{\alpha}^{\beta} radius(arc\ \ length)$

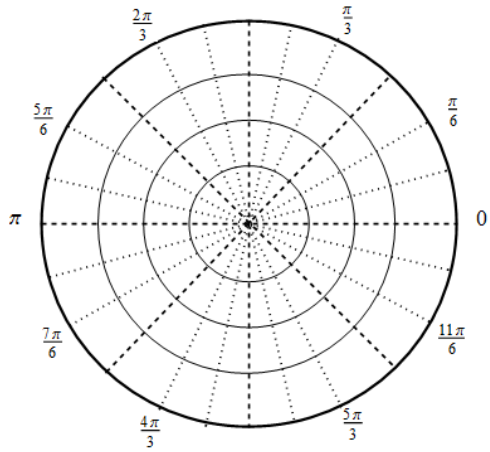
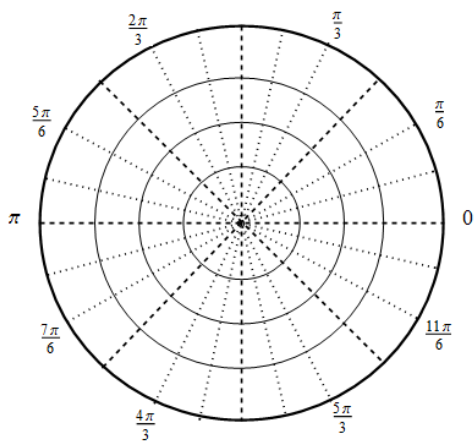
5. a) Find the Cartesian coordinates for the polar coordinate $(-3, -3\pi/4)$

b) Find polar coordinates for the Cartesian coordinate $(2, -\sqrt{2})$ where $r < 0, and\ \theta > 0$

6. Sketch the graph. $r = 2 + 2\sin 2\theta$.



7. Sketch the graph. $r = 3 + 2\cos\theta$



8. Sketch the graph. $r = 2 + 3\cos\theta$